Explicit Constraint Decisions for Differential Polynomials over the Rational Constants

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Abstract

The constrained-pair predicate asks whether every two solutions of a differential equation, after excluding a lower-order zero set, generate isomorphic differential fields with the designated solutions identified. We give exact decisions for two classes over the rational field with zero derivation. For an irreducible first-order polynomial $p(Y, DY)$, put $r(Y) = p(Y, 0)$. If $r \neq 0$, a nonzero polynomial $q(Y)$ is a constraint exactly when the squarefree part of r divides q; if $r = 0$, none is. For the linear equation $f(D)Y = 0$, where the roots of the monic rational polynomial f are linearly independent over $\mathbb{Q}$, the constraint criterion is divisibility by an explicit rational product of spectral linear forms. A resultant computes this product without manipulating the roots. Cyclotomic examples give an elementary recognizable family of unbounded differential orders. We prove both directions of each criterion and construct the needed solutions algebraically. These are restricted decisions for the actual constrained-pair predicate, not a decision procedure for all differential polynomials over $\mathbb{Q}$.

Contents

1 A differential analogue of irreducibility **1**

2 Definitions and algebraic preliminaries **2**

3 Every first-order input in the stated slice **3**

4 Higher-order equations with independent spectral values **4**

5 A rational algorithm and explicit unbounded examples **6**

6 Why dependent spectra require a different argument **7**

7 Relation to the general constraint-set problem **8**

1 A differential analogue of irreducibility

For an ordinary irreducible polynomial over $\mathbb{Q}$, any two roots generate fields isomorphic over $\mathbb{Q}$ by an isomorphism sending one root to the other. A differential equation is more complicated. Two

solutions of the same equation may satisfy different additional differential equations and need not generate the same kind of differential field. A constraint excludes lower-order exceptional solutions until a single designated-root isomorphism type remains.

Miller, Ovchinnikov and Trushin [1] study the computational role of these pairs as an analogue of irreducible polynomials. Their broad question includes the computability of constraint sets. Even with base $\mathbb{Q}$ and zero derivation, a general procedure is not supplied by their results. Here we treat two concrete input classes. In each case the problem reduces to ordinary rational polynomial divisibility.

The two mechanisms are different. In order one, all nonconstant solutions of an irreducible equation have the same differential function field; one must exclude its constant solutions. In higher order, spectral independence forces all nonzero spectral coordinates to be algebraically independent; one must exclude solutions missing a coordinate. This distinction also explains why the first-order argument cannot simply be iterated.

2 Definitions and algebraic preliminaries

A differential field is a field F with a derivation D, satisfying $D(a+b) = Da + Db$ and $D(ab) = aDb + bDa$. All fields here have characteristic zero, there is one derivation, and $\mathbb{Q}$ has its zero derivation. Write

$$\mathbb{Q}\{Y\} = \mathbb{Q}[Y_0, Y_1, Y_2, \ldots], \qquad DY_i = Y_{i+1}.$$

Evaluation at a sends Y_i to $D^i a$. The order of a nonconstant differential polynomial is the largest derivative index it uses; nonzero constants are treated as having order below zero. Write $\mathbb{Q}\langle a\rangle$ for the differential field generated by a.

Definition 2.1. Let p be a source-normalized, algebraically irreducible differential polynomial of order n, and let $q \neq 0$ have order less than n. The pair (p, q) is *constrained over* $\mathbb{Q}$ if there is a solution of $p(a) = 0$, $q(a) \neq 0$ in a differential extension and any two such solutions a, b in arbitrary differential extensions admit a differential-field isomorphism

$$\mathbb{Q}\langle a\rangle \longrightarrow \mathbb{Q}\langle b\rangle, \qquad a \longmapsto b,$$

fixing $\mathbb{Q}$. Denote the positive predicate by $\mathrm{Con}_{\mathbb{Q}}(p, q)$.

This is the designated-root condition of [1, Definition 4.3]. Their notation $T_{\mathbb{Q}}$ is the complementary predicate, so signs must be kept distinct when translating statements. Scalar normalization of p does not change its solutions. In the first-order theorem we assume that source normalization, not the stronger condition that the coefficient of the highest derivative be 1.

Lemma 2.2. *An element algebraic over $\mathbb{Q}$ in a differential extension is a constant. A nonconstant element is therefore transcendental over $\mathbb{Q}$.*

Proof. If $m(a) = 0$ is its irreducible rational polynomial, differentiation gives $m'(a)Da = 0$. Separability in characteristic zero gives $m'(a) \neq 0$, hence $Da = 0$. □

We also use the elementary fact that a derivation on a characteristic-zero field extends uniquely to an algebraic extension. For an algebraic element a with minimal polynomial $\sum_i c_i T^i$, its required derivative is

$$Da = -\frac{\sum_i (Dc_i)a^i}{\sum_i i c_i a^{i-1}}.$$

The denominator is nonzero by separability. This formula extends the derivation successively to finite algebraic extensions and then to an algebraic closure; uniqueness makes the extensions compatible. In particular every algebraic number remains constant.

3 Every first-order input in the stated slice

Let $p(Y, Z) \in \mathbb{Q}[Y, Z]$ be algebraically irreducible and have positive degree in Z, representing $Z = DY$. Let $0 \neq q(Y) \in \mathbb{Q}[Y]$, and put $r(Y) = p(Y, 0)$. For $r \neq 0$, let $\mathrm{rad}(r)$ be its monic squarefree part, with $\mathrm{rad}(r) = 1$ for nonzero constant r.

Theorem 3.1. *On these source-valid inputs:*

1. *if $r \neq 0$, then $\mathrm{Con}_{\mathbb{Q}}(p, q)$ holds exactly when $\mathrm{rad}(r) \mid q$ in $\mathbb{Q}[Y]$;*
2. *if $r = 0$, then $\mathrm{Con}_{\mathbb{Q}}(p, q)$ fails for every nonzero q.*

3.1 The common field of all nonconstant solutions

Lemma 3.2. *Any two nonconstant solutions of $p(a, Da) = 0$ generate differential fields isomorphic over $\mathbb{Q}$ by a map sending one solution to the other.*

Proof. Such an a is transcendental by Lemma 2.2. The polynomial p is primitive as a polynomial in Z over $\mathbb{Q}[Y]$: otherwise its nonconstant content would factor it. Gauss's lemma makes it irreducible over $\mathbb{Q}(Y)$. Substitution $Y \mapsto a$ therefore makes $p(a, Z)$ the minimal polynomial of Da, up to a nonzero scalar over $\mathbb{Q}(a)$. It follows that the evaluation kernel of $\mathbb{Q}[Y, Z] \to \mathbb{Q}[a, Da]$ is exactly (p). One can see the contraction directly: an element of $\mathbb{Q}[Y, Z]$ divisible by primitive p in $\mathbb{Q}(Y)[Z]$ is divisible in $\mathbb{Q}[Y, Z]$.

The partial derivative p_Z is nonzero and has smaller Z-degree than p. It is not in (p), so $p_Z(a, Da) \neq 0$. Differentiating the equation gives

$$D^2 a = -\frac{p_Y(a, Da)Da}{p_Z(a, Da)}. \tag{3.1}$$

Thus $F_a = \mathbb{Q}(a, Da)$ is closed under the derivation, and equals $\mathbb{Q}\langle a\rangle$. For another nonconstant solution b, the two evaluation maps identify both fields with $\mathrm{Frac}(\mathbb{Q}[Y, Z]/(p))$. The resulting isomorphism sends a, Da to b, Db. It commutes with D on the first generator by construction, on the second by (3.1), and hence on the whole field by the derivation rules. □

The separant $p_Z(a, Da)$ is therefore nonzero at every nonconstant solution in this setting. It is a conclusion of irreducibility and transcendence, not a hidden genericity assumption.

Lemma 3.3. *If $r \neq 0$, there exists a nonconstant solution avoiding every specified nonzero $q(Y)$.*

Proof. In the domain $\mathbb{Q}[Y, Z]/(p)$ let y, z be the residue classes, and let F be its fraction field. The class y is transcendental over $\mathbb{Q}$, and $p_Z(y, z) \neq 0$. Set

$$Dy = z, \qquad Dz = -p_Y(y, z)z/p_Z(y, z).$$

These prescriptions respect the defining relation because $D(p(y,z)) = 0$, so they define a derivation on the fraction field. Equivalently extend the usual formal derivation on the localized polynomial ring and pass through the relation. Since $r \neq 0$, irreducible p is not a scalar multiple of Z, so $z \neq 0$. Thus y is nonconstant, solves $p(y, Dy) = 0$, and avoids q by transcendence. This is an explicit algebraic differential extension of $\mathbb{Q}$; no analytic solution or differentially closed field is needed. □

3.2 Removing precisely the constant exceptions

Proof of Theorem 3.1. Assume $r \neq 0$. Constant solutions are exactly the algebraic constants with $r(a) = 0$. Indeed a constant solution satisfies that equation; conversely every root of r in an algebraic closure of $\mathbb{Q}$, equipped with the zero derivation, is a solution. A nonzero rational polynomial vanishes at all these roots precisely when it is divisible by $\mathrm{rad}(r)$.

If $\mathrm{rad}(r) \mid q$, any solution avoiding q is therefore nonconstant. Lemma 3.2 gives the required isomorphism between any two, and Lemma 3.3 supplies existence.

If $\mathrm{rad}(r) \nmid q$, take an irreducible factor of r not dividing q and one of its algebraic roots c. Then $q(c) \neq 0$ and c is a constant solution. The nonconstant solution from Lemma 3.3 also avoids q. A differential isomorphism sending one to the other would preserve vanishing of DY, which is impossible. Hence the pair is not constrained.

If $r = 0$, then Z divides p. Irreducibility forces p to be a nonzero rational multiple of Z. Infinitely many rational constants avoid q. Two distinct such constants are solutions but cannot be mapped to each other by an isomorphism fixing $\mathbb{Q}$. This proves the second case. □

Corollary 3.4. *The constrained-pair predicate is decidable on this first-order slice.*

Proof. Compute r by substitution. If zero, answer no. Otherwise compute $\gcd(r, r')$, form the monic associate of $r/\gcd(r, r')$, and test its divisibility into q. These are finite rational polynomial operations and implement the theorem. □

Example 3.5. For $p = Z - Y$, $r = -Y$, so exactly the nonzero q divisible by Y are constraints. The equation has the exceptional solution zero; every nonzero solution has the same designated-root differential field. For $p = YZ - 1$, $r = -1$, so every nonzero $q(Y)$ works. For $p = Z$, none works, since lower-order exclusion by one nonzero polynomial cannot reduce all rational constants to a single designated value.

At order zero, source-valid irreducible $p(Y)$ and nonzero constant q are constrained for the usual algebraic reason: all roots are algebraic constants and their ordinary $\mathbb{Q}$-isomorphisms are differential isomorphisms. This small case should not be confused with the first-order equation $DY = 0$.

4 Higher-order equations with independent spectral values

Let

$$f(T) = T^n + a_{n-1}T^{n-1} + \cdots + a_0 \in \mathbb{Q}[T], \qquad n \geq 1,$$

and suppose its roots $\lambda_1, \dots, \lambda_n$ in a splitting field K are linearly independent over $\mathbb{Q}$. In particular they are nonzero and pairwise distinct. Put

$$p(Y) = f(D)Y = Y_n + a_{n-1}Y_{n-1} + \cdots + a_0 Y_0.$$

This differential polynomial is algebraically irreducible because it is monic linear in Y_n, whether or not f is irreducible as an ordinary univariate polynomial. Let $0 \neq q \in \mathbb{Q}[Y_0, \dots, Y_{n-1}]$.

For each i, define the linear form

$$L_i(Y) = \left[\prod_{j \neq i}(D - \lambda_j)\right] Y, \qquad R_f(Y) = \prod_{i=1}^{n} L_i(Y).$$

The Galois group permutes these forms, so their product has rational coefficients. It uses only the first n jets and is nonzero.

Theorem 4.1. *Under this spectral independence hypothesis,*

$$\mathrm{Con}_{\mathbb{Q}}(p, q) \quad \Longleftrightarrow \quad R_f \mid q \quad \text{in } \mathbb{Q}[Y_0, \dots, Y_{n-1}].$$

In particular (p, R_f) is a positive constrained pair.

4.1 Spectral coordinates are valid for every solution

Let a be any solution in any differential extension. Extend its field algebraically so that it contains a copy of K. Its elements are constants by Lemma 2.2. Define

$$x_i = L_i(a)/f'(\lambda_i).$$

Lagrange interpolation gives, for $0 \leq k < n$,

$$D^k a = \sum_i \lambda_i^k x_i, \qquad Dx_i = \lambda_i x_i. \tag{4.1}$$

For the first identity, interpolate T^k at the distinct roots using the polynomials $\prod_{j \neq i}(T - \lambda_j)/f'(\lambda_i)$, then apply the resulting operator identity to a. For the second use $(D - \lambda_i)L_i(a) = p(a) = 0$. The Vandermonde matrix in (4.1) is invertible, so these are invertible linear coordinates on the jets, and

$$R_f(a) = \left(\prod_i f'(\lambda_i)\right) \prod_i x_i. \tag{4.2}$$

Lemma 4.2. *If all x_i are nonzero and $Dx_i = \lambda_i x_i$, they are algebraically independent over K.*

Proof. For a multi-index $\alpha \in \mathbb{N}^n$, the monomial x^α has derivative $w_\alpha x^\alpha$, with $w_\alpha = \sum_i \alpha_i \lambda_i$. Rational linear independence of the λ_i makes these weights distinct for distinct multi-indices.

Suppose a nonzero polynomial relation $\sum_{\alpha \in S} c_\alpha x^\alpha = 0$ holds, after combining equal monomials and discarding zero coefficients. Fix $\alpha_0 \in S$ and apply the differential operator $\prod_{\beta \in S \setminus \{\alpha_0\}}(D - w_\beta)$. The coefficients and weights are constants. Every term except α_0 vanishes, leaving

$$c_{\alpha_0} \prod_{\beta \neq \alpha_0} (w_{\alpha_0} - w_\beta) x^{\alpha_0} = 0.$$

All factors are nonzero in a field, a contradiction. This also handles a constant monomial if the relation has one. □

Lemma 4.3. *If $p(a) = 0$ and $R_f(a) \neq 0$, its first n jets are algebraically independent over $\mathbb{Q}$. Any two such solutions have the required designated-root differential isomorphism over $\mathbb{Q}$.*

Proof. By (4.2), all spectral coordinates are nonzero. Lemma 4.2 and the invertible change of coordinates give jet independence even over K. The recurrence $p(a) = 0$ expresses $D^n a$ as the fixed rational linear combination of lower jets. Thus $F_a = \mathbb{Q}(a, Da, \ldots, D^{n-1}a)$ is closed under D and equals $\mathbb{Q}\langle a\rangle$. For a second solution b, algebraic independence gives an isomorphism sending each $D^k a$ to $D^k b$. It commutes with the derivation on the first $n-1$ generators by construction and on the last by the common recurrence. Hence it is a differential isomorphism on the whole field, fixing $\mathbb{Q}$ and sending a to b. □

4.2 Generic and missing-coordinate solutions

Use algebraically independent variables $X_1, \ldots, X_n$ over K and give the rational function field $K(X_1, \ldots, X_n)$ the derivation $DX_i = \lambda_i X_i$, zero on K. This is a valid differential field by the polynomial and quotient rules. The element $b = \sum_i X_i$ solves $p(b) = 0$. Its jets are the invertible Vandermonde transform of the X_i, so every nonzero rational polynomial q in those jets is nonzero at b. Also $R_f(b) \neq 0$. This supplies the existence needed in Definition 2.1.

If one sets $X_i = 0$ and keeps the other variables independent, the smaller rational function field $K(X_j : j \neq i)$, with the same derivation rule, gives a solution $c = \sum_{j \neq i} X_j$ with a missing coordinate. It satisfies $R_f(c) = 0$.

Proof of Theorem 4.1. If $R_f \mid q$, a solution avoiding q has $R_f \neq 0$. Lemma 4.3 gives the desired isomorphism between any two, and the all-coordinate construction supplies existence.

Suppose $R_f \nmid q$. Divisibility does not change on extending coefficients to K: if $q = R_f h$ over K, Galois invariance and uniqueness of the quotient imply $h \in \mathbb{Q}[Y_0, \ldots, Y_{n-1}]$. The L_i are pairwise nonassociate linear coordinate forms over K. Since their product does not divide q, some L_i does not divide q. In spectral coordinates, the restriction of q to $X_i = 0$ is therefore nonzero.

Choose the missing-coordinate solution c just constructed. Its remaining coordinates are independent, so $q(c) \neq 0$, while $R_f(c) = 0$. Compare it with the all-coordinate solution b, for which both $q(b)$ and $R_f(b)$ are nonzero. The differential polynomial R_f over $\mathbb{Q}$ distinguishes them, so no designated-root differential isomorphism exists. For $n = 1$ the missing-coordinate field is simply K and $c = 0$; the same argument applies. This proves necessity. □

5 A rational algorithm and explicit unbounded examples

The proof used a splitting field, but the decision polynomial can be computed over $\mathbb{Q}$ alone. Write $a_n = 1$ and set

$$G_f(T) = \sum_{k=1}^{n} a_k \sum_{j=0}^{k-1} T^j Y_{k-1-j}.$$

It is obtained from $(f(S) - f(T))/(S - T)$ by replacing each S^m with Y_m. Substituting $T = \lambda_i$ gives L_i, so

$$R_f = \operatorname{Res}_T(f(T), G_f(T)). \tag{5.1}$$

Here the resultant identity uses that f is monic. A Sylvester determinant computes this polynomial in rational arithmetic. Multivariate polynomial division by the single polynomial R_f then decides whether $R_f \mid q$.

This is a terminating decision algorithm on the spectral promise. It does not need the roots as additional input. No implementation of a general promise recognizer is asserted. The following subclass has an elementary recognizable promise.

Example 5.1 (A quadratic equation)**.** Take $f(T) = T^2 - T - 1$. Its roots $(1+\sqrt{5})/2, (1-\sqrt{5})/2$ are rationally independent: the rational and $\sqrt{5}$ parts of a rational relation force both coefficients to vanish. Then

$$p = Y_2 - Y_1 - Y_0, \qquad R_f = Y_1^2 - Y_0 Y_1 - Y_0^2.$$

Every nonzero $q(Y_0, Y_1)$ is decided: the pair is constrained exactly when this displayed quadratic divides q. In particular $q = R_f$ gives a positive pair and $q = 1$ a negative one.

Proposition 5.2. *For every prime ℓ, the polynomial $f(T) = 1+T+\cdots+T^{\ell-1}$ satisfies the spectral promise. Consequently the theorem supplies positive pairs and full decisions for all nonzero q at an unbounded set of differential orders.*

Proof. The roots are the nontrivial ℓth roots $\zeta, \ldots, \zeta^{\ell-1}$. The polynomial f is irreducible: substitute $T = U + 1$ and apply Eisenstein at ℓ to $((U+1)^\ell - 1)/U$. If a rational relation among the displayed roots existed, it would be a polynomial of degree at most $\ell - 1$ with zero constant term vanishing at ζ. Its minimal polynomial is f, so this relation polynomial would be a scalar multiple of f. The zero constant term forces that scalar to be zero. The roots are therefore rationally independent. The orders are $\ell - 1$, unbounded as primes grow. Trial division and checking the coefficients recognize this family. □

6 Why dependent spectra require a different argument

Spectral independence is not a disposable sufficient condition. If the weights are dependent, nonzero coordinates can carry additional rational invariants, distinguishing solutions which all avoid the proposed product.

For $f(T) = T^2 - 1$, the roots $1, -1$ are dependent, and the analogous product is $R = Y_1^2 - Y_0^2$. In $\mathbb{Q}(x)$ with $Dx = x$, put

$$a_c = (x + c/x)/2 \qquad (c \in \mathbb{Q} \setminus \{0\}).$$

Then $D^2 a_c = a_c$ and $R(a_c) = -c$. The choices $c = 1, 2$ both avoid R but are distinguished by $R + 1$, which vanishes at only the first. Thus even $(Y_2 - Y_0, R)$ is not constrained. The independent-spectrum theorem would give a false answer here if its hypothesis were omitted.

At order one the exceptional solutions were precisely the constants. At higher order a solution may have several nonzero derivatives and still satisfy additional algebraic or differential relations among its jets. The spectral proof succeeds by proving jet independence, not merely by observing nonconstancy.

7 Relation to the general constraint-set problem

The first theorem decides every supplied nonzero lower-order q for the specified irreducible order-one inputs over the pure rational constant field. The second decides every supplied nonzero q for an unbounded-order family. These are decisions for the designated-root predicate itself, including both existence and all-extension isomorphism requirements. They do not replace $\mathbb{Q}$ by a field with a named nonconstant parameter or ask only whether some constraint exists.

The proofs use standard differential-field and polynomial methods; the local contribution is the explicit criterion and its effective form for these inputs. The general predicate for repeated roots, dependent spectra and nonlinear higher-order equations is not decided here. Neither theorem establishes the broader differential Kronecker or extension-property conclusions discussed in [1].

A bounded source and literature check did not identify an equivalent published formulation of both criteria. This is not a determination of historical novelty; the first-order function-field observation and the spectral independence argument in particular belong to classical differential algebra. Further attribution work is appropriate before claiming these as new literature results.

Preparation. This AI-assisted hand-proof manuscript includes all local algebraic constructions and both directions of the decision criteria. The algorithms are proved terminating on their stated input promises; no empirical verification or formal certification is claimed.

References

[1] R. Miller, A. Ovchinnikov and D. Trushin, *Computing Constraint Sets for Differential Fields*, Journal of Algebra 407 (2014), 316–357. doi:10.1016/j.jalgebra.2014.02.032. Source version arXiv:1208.1152v2, 12 April 2014; especially Definition 4.3 and the constraint-set decidability discussion.